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electric flux through the closed surface of the sphere using the equation (1.58).

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \oint E\, dA \cos\theta$$

The electric field of the point charge is directed radially outward at all points on the surface of the sphere. Therefore, the direction of the area element $d\vec{A}$ is along the electric field $\vec{E}$ and $\theta = 0°$.

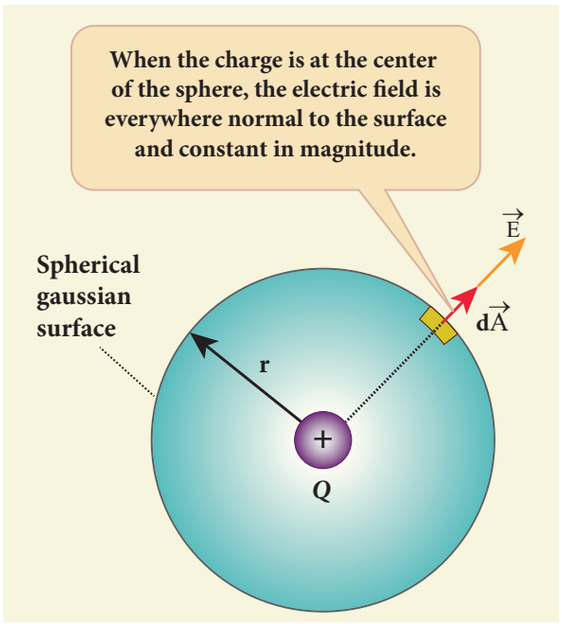


Figure 1.36 Total electric flux of point charge

$$\Phi_E = \oint E\, dA \qquad \text{since } \cos 0^0 = 1 \qquad (1.59)$$

E is uniform on the surface of the sphere,

$$\Phi_E = E \oint dA \qquad (1.60)$$

Substituting for $\oint dA = 4\pi r^2$ and $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$ in equation 1.60, we get

$$\Phi_E = \frac{1}{4\pi\varepsilon_\circ}\frac{Q}{r^2} \times 4\pi r^2 = 4\pi \frac{1}{4\pi\varepsilon_\circ} Q$$

$$\Phi_E = \frac{Q}{\varepsilon_\circ} \qquad (1.61)$$

The equation (1.61) is called as Gauss's law. The remarkable point about this result is that the equation (1.61) is equally true for any arbitrary shaped surface which encloses the charge Q and as shown in the Figure 1.37. It is seen that the total electric flux is the same for closed surfaces A_1, A_2 and A_3 as shown in the Figure 1.37.

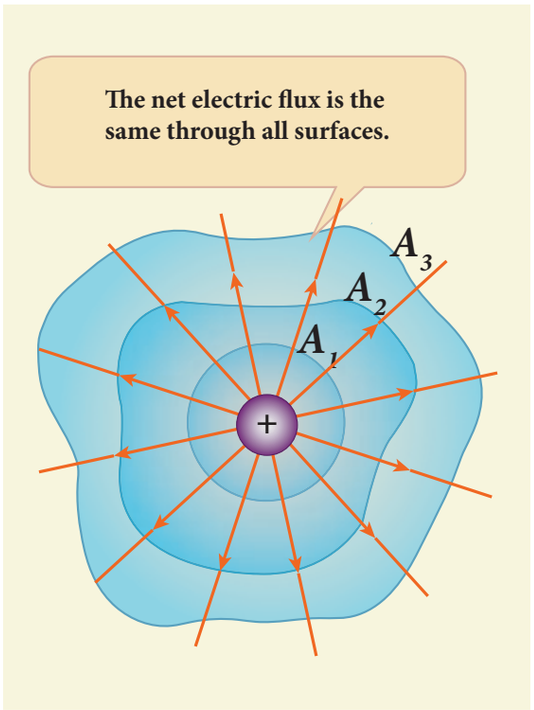


Figure 1.37 Gauss law for arbitrarily shaped surface

Gauss's law states that **if a charge Q is enclosed by an arbitrary closed surface, then the total electric flux Φ_E through the closed surface is**

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\varepsilon_\circ} \qquad (1.62)$$

Here Q_{encl} denotes the charges inside the closed surface.

Discussion of Gauss law

(i) The total electric flux through the closed surface depends only on the

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